

Structure-Based UAV Visual Geo-Localization via Delaunay Triangulation and Ekeland Free-Cone Angle Descriptors

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Abstract

Accurate geo-localization of unmanned aerial vehicles (UAVs) in GPS-denied environments remains a core challenge for autonomous navigation. We study a scene-agnostic, training-free approach that seeks the UAV’s 2-D geographic position by matching purely geometric building-footprint descriptors extracted from a single nadir image against a geo-referenced satellite map, and we provide a rigorous analysis of where and why this appealing paradigm succeeds and fails. Buildings are segmented by a pre-trained Mask R-CNN; each polygon is decomposed into triangles by Constrained Delaunay Triangulation (CDT), and each triangle is encoded with a 5-dimensional descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$ comprising two sorted interior angles and three **Maximal Free Cone Angles (MFCA)**—a geometric quantity measuring the largest obstacle-free angular sector at each vertex. The MFCA is directly inspired by Ekeland’s variational principle (1974): the principle guarantees a critical point where no improvement is achievable without a proportional distance cost; we instantiate this as a *kernel expansion* P_{sub} —the largest polygon formed by merging adjacent CDT triangles such that all three original triangle vertices remain on the boundary of the result. This stopping criterion is the direct geometric analogue of Ekeland’s cone-criticality condition: the polygon cannot be expanded further while keeping the triangle vertices as boundary (critical) points. The Ekeland angle e_v at each vertex is then the exterior angle of P_{sub} at v , computed in $O(1)$ per vertex by an edge-anchored half-plane construction with no sweep required. Satellite patch descriptors are indexed offline in a KD-tree (ℓ_1 metric); online matching runs in $O(M \log N)$ time per query. We evaluate this pipeline rigorously on the UAV-VisLoc benchmark and report a central negative result with a precise diagnosis. Building coverage at the bench location is high (93%), yet single-frame descriptor retrieval fails at city scale: the geometrically correct match is essentially never among the top-ranked candidates (retrieval $<1\%$) and localization error remains at the kilometre level. This ceiling is one of descriptor *discriminability*, not of coverage or feature scaling—enriching the descriptor with building size and a log-polar building-constellation context, matching at building rather than triangle granularity, and normalizing feature dimensions each shrink the true-match descriptor distance by up to $16\times$ yet leave retrieval near zero. Following the coarse-to-fine, particle-filter paradigm of successful vector-map methods, we further integrate the per-frame matches over a flight trajectory with a Gaussian-mixture observation and a particle filter; this too fails to converge, because the true position receives recurring observation support in only 12% of frames. This limit is terrain-general: across five UAV-VisLoc sites spanning urban, coastal, hilly, built-up, and arid terrain, meter-level accuracy stays below 3% within 500 m with kilometre-scale median error. We conclude that the limiting factor is the *observation model*: purely geometric per-element matching does not supply the recurring, spatially consistent support that sequential estimation requires at scale, and must be paired with a stronger area-level observation. Acting on this diagnosis, we show the fix is constructive and keeps the Ekeland angle central: aggregating the descriptor over ≈ 200 m cells—as a histogram of per-building Ekeland free-cone angles alongside building density and layout—escapes the per-element ceiling. Against real re-implemented baselines (BRM, CFBVM’s concentric-ring descriptor) under identical protocol, this

pooled, statistically significant improvement in coarse-localization recall (+2.9–4.4 percentage points, $p \leq 0.02$, $n=272$ frames across five sites) is not universal per site—the Ekeland angle is the best performer on three of five terrains. The Ekeland free-cone angle is thus a complementary, discriminative shape cue at the area level on most, not all, terrains, though absolute accuracy remains coarse. Our analysis provides the missing empirical account of why training-free geometric matching, though attractive, does not by itself scale—and where the Ekeland descriptor nonetheless adds measurable value.

1 Introduction

UAVs depend on GNSS for position fixes in nearly every operational domain: search-and-rescue, precision agriculture, infrastructure inspection, and military reconnaissance. Yet GNSS is unavailable in urban canyons, jammed in contested airspace, and unreliable indoors—scenarios that are precisely those requiring the most reliable autonomy. A single GNSS outage during a search-and-rescue operation can redirect a UAV kilometres from the target zone. In contested airspace, GPS jamming has been documented to deny localization entirely for extended periods [11, 26]. For emergency responders, infrastructure inspectors, and autonomous delivery operators, GNSS failure is not a theoretical edge case—it is the operational norm in precisely the environments these platforms are deployed. Vision-based absolute visual localization (AVL) addresses this gap by recovering the UAV’s 2-D geographic position from onboard camera images matched against a pre-stored reference map, with no satellite signal required [6, 29].

The dominant difficulty in UAV-to-satellite AVL is the *cross-view appearance gap*: satellite maps are orthorectified nadir projections captured in a fixed season, while UAV images arrive at varying altitude, heading, roll, pitch, time of day, and season. Existing methods expose a fundamental trilemma: (i) *Appearance-based methods* (SIFT, ORB, deep cross-view networks [13, 15]) rely on pixel-level signals that change across season, altitude, and lighting—SIFT achieves zero Recall@10 on MAFS-10 after 1,630 s of computation [35], confirming that texture descriptors cannot bridge the nadir gap. (ii) *Deep learning methods* overcome the appearance gap by learning invariant embeddings, but require scene-specific retraining, making them impossible to deploy on a new city map within hours. (iii) *Vector map methods* exploit shape rather than texture, but existing approaches (BRM [4], CFBVM [29]) operate on whole-building statistics and lose the *spatial context between buildings*, making them poorly discriminative in dense urban blocks where many buildings share similar individual shapes. No existing training-free method simultaneously encodes both triangle-level shape and the structural context of the surrounding building footprint within a single descriptor.

Our key insight is that *building footprint geometry* is stable across all of these appearance variations: a polygon extracted from a nadir UAV image and the same polygon in a satellite map agree in shape regardless of altitude, season, or illumination. We exploit this by (i) decomposing each building polygon into triangles via CDT and (ii) encoding each triangle with a 5-D descriptor of pure angular quantities invariant to scale, rotation, and appearance. The free-space component of this descriptor—the Maximal Free Cone Angle (MFCA)—is directly inspired by Ekeland’s variational principle [12]: the principle characterises a point of maximal local stability as the apex of the largest cone that touches but does not cross a function’s graph; we instantiate this cone not over a function graph but over a 2-D polygon boundary, yielding a scalar that encodes *how much local geometric freedom exists at each building vertex*.

Training-free geometric localization is an appealing route to *map-portable autonomy*—a platform pre-loaded with satellite imagery of any new city operating immediately, with no training phase. This paper asks, and answers rigorously, whether purely geometric per-element matching can deliver it: we find that it cannot at city scale, and we characterize precisely why, turning an aspirational claim into a measured account of the paradigm’s limits and of what a viable observation model must provide.

Contributions.

1. **Edge-anchored Ekeland angle (MFCA).** We derive the Ekeland angle e_v from first principles via a two-phase construction: (i) expand the kernel triangle into a maximal polygon P_{sub}

whose stopping condition is geometrically analogous to Ekeland’s cone-criticality—merging halts precisely when any of the three original triangle vertices would leave the boundary; (ii) compute e_v as the exterior angle of P_{sub} at v via an edge-anchored half-plane selection (Section 5.4). The result is *exact* (no quantization), $O(1)$ per vertex, and geometrically motivated: the kept sector’s bounding ray is tangent to the polygon boundary, the same picture as Ekeland’s “cone touches but does not cross” condition.

2. **5-D geometric feature and matching pipeline.** A descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$ combining two sorted interior angles and three sorted MFCAs (Section 5.4), with a complete training-free pipeline—instance segmentation, CDT, KD-tree index (ℓ_1 , $O(M \log N)$ online matching), and a translation-consensus position estimator (Sections 5.5–5.6).
3. **A rigorous discriminability study on UAV-VisLoc.** Using a truth-anchored retrieval diagnostic, we show that per-element geometric matching hits a *discriminability ceiling* at city scale (retrieval 0% in the top-20 despite 93% coverage), that it is robust to a battery of descriptor enrichments (size, constellation context, building-level granularity, normalization—each shrinking the true-match distance up to $16\times$ without effect), and that a CFBVM-style particle filter cannot recover it because the true position is supported in only 12% of frames. The ceiling is terrain-general: it holds across five UAV-VisLoc sites spanning urban, coastal, hilly, built-up, and arid terrain (Section 6.3). We conclude that the limiting factor is the observation model, and identify what a scale-robust one must supply (Sections 6.5–6.8).
4. **An area-level Ekeland descriptor that recovers discriminability.** Acting on the diagnosis, we aggregate the descriptor over ≈ 200 m cells as a histogram of per-building Ekeland free-cone angles (with density and layout). This escapes the per-element ceiling and, against real re-implemented BRM and CFBVM baselines under identical protocol, significantly improves pooled coarse-localization recall across five sites ($n=272$, $p \leq 0.02$)—though the effect is not universal per site (best on 3 of 5 terrains)—isolating the Ekeland angle as a complementary, discriminative shape cue at the area level, with absolute accuracy remaining coarse (Section 6.12).

Scope and assumptions. The method assumes (A1) the UAV camera points approximately nadir (tilt $< 15^\circ$, corrected by IMU), (A2) heading ϕ is available from onboard compass/IMU to within $\pm 5^\circ$ (standard for consumer autopilots), and (A3) buildings are present in the field of view. The descriptor and matching stages require *no training data specific to the target scene*; only the Mask R-CNN backbone is pre-trained on public aerial building datasets [14, 17]. We analyze the pipeline’s behavior—including the role of building coverage and the discriminability of the geometric descriptor—experimentally (Section 6.5).

Broader impact. A lightweight, training-free geo-localizer could benefit emergency response, infrastructure inspection, and autonomous navigation in GPS-denied or communication-restricted environments. We acknowledge that the same capability carries dual-use implications in contested or surveillance scenarios; we encourage responsible deployment with appropriate access controls.

2 Related Work

Classical template and feature matching. Template matching methods (NCC, Phase Correlation, Mutual Information) compare query images to raster satellite tiles by direct pixel similarity. NCC-based template matching is reported to require hundreds of seconds per query and collapses under seasonal illumination shifts [29]. SIFT [24], SURF [2], and ORB [30] extract keypoints robust to scale and rotation, yet the nadir-to-satellite appearance gap overwhelms these features: SIFT achieves 62.8 m RMSE with zero Recall@10 on MAFS-10, while ORB achieves 21.1 m with only 7.9% Recall@10 [35]. The abBRIEF descriptor embedded in Monte Carlo Localization reduces error to ≈ 17.8 m but is constrained to $\sim 1 \text{ km}^2$ coverage [25].

Polygon, shape, and visibility descriptors. Classical polygon descriptors offer a natural alternative for VMAP-based localization. Turning functions [1] represent polygon shape as cumulative turning

angle along the boundary—invariant to scale and rotation but sensitive to segmentation noise and vertex ordering. Shape contexts [3] encode the distribution of boundary points in a log-polar histogram around each vertex, offering partial deformation invariance but requiring dense correspondences and scaling poorly to large databases. Polygon angle histograms aggregate interior angle distributions per building instance, yielding a compact per-building signature but losing the spatial inter-triangle context that discriminates geometrically similar buildings in different surroundings.

Our MFCA is conceptually related to the *visibility polygon* and *largest empty sector* problems from computational geometry [9, 28]: given a query point inside a polygonal environment, the visibility polygon enumerates all directions not obstructed by boundary segments. MFCA is a coarser scalar summary—the exterior angle at each vertex of the kernel expansion polygon P_{sub} —but encodes building-shape context through the expansion process, making it discriminative across different polygon geometries, not just visible directions. Crucially, our construction is motivated by Ekeland’s variational principle [12] rather than being an ad hoc design choice: the stopping criterion of the kernel expansion (original triangle vertices must remain on the boundary) is geometrically analogous to the Ekeland criticality condition, and the edge-anchored angle realises the “cone touches but does not cross” picture directly, via a single closed-form expression (Definition 5.3). The result is an $O(1)$ -per-vertex exact computation, with no search radius and no angular sweep, that was previously approximated by a 1° -sweep. This grounding distinguishes MFCA from visibility-polygon summaries, which are defined by geometric coverage alone rather than by a variational stability criterion.

Our descriptor operates on *triangles within a building*, not whole-polygon statistics, and encodes building-shape context through the edge-anchored Ekeland angle—a structural quantity unavailable to any single-polygon descriptor.

Semantic vector map matching. Vector map (VMAP) methods avoid raster storage and cross-modal appearance gaps. Choi & Myung [4] proposed Building Ratio Map (BRM), matching building pixel-density ratios per grid cell against OpenStreetMap; it achieves 7.5–12 m RMSE in a 5.7 km² area but discards shape information and degrades at large scale. Nassar et al. [27] applied deep CNN segmentation and shape-based matching, reporting 6.3–10.4 m average drift across two flight datasets (Potsdam, Famagusta). Ouyang et al. [29] proposed Coarse-to-Fine Building Vector Matching (CFBVM) combined with a GMM particle filter, reporting 6.6 m mean error over 11,025 km² using a shape vector defined over concentric density rings (as reported in the published abstract/summary; we could not access the full text to verify experimental detail independently). Most closely related to our positive result, Wang et al. [32] match buildings jointly on *shape and spatial relationship* (SSRM) against clean vector (‘.shp’) e-maps, explicitly motivated by the observation—which our analysis independently quantifies—that per-building *shape matching alone cannot capture building uniqueness*. A key distinction is the map source: SSRM matches against precise vector footprints, whereas we match UAV-segmented footprints against *satellite-segmented* ones, where cross-view segmentation disagreement (Section 6.5) makes exact per-building correspondence infeasible—which is why our discriminability gain appears at the area level rather than per building. Our method shares the VMAP paradigm with CFBVM but substitutes ring-area summation with a triangle-level angular descriptor that requires no overlap area computation, is invariant to segmentation polygon ordering, and carries explicit free-space context through the kernel-expansion MFCA; unlike CFBVM’s shape-agnostic density rings, its area-level form retains per-building shape via Ekeland-angle histograms (Section 6.12).

Deep learning and sequential methods. End-to-end cross-view networks (FPI [7], SSPT [13], Goforth & Lucey [15]) learn appearance-invariant embeddings but require scene-specific retraining to generalize. The Semantic-Weighted Adaptive Particle Filter (SWA-PF) [35] combines SegFormer [33] semantics with 4-DoF particle filtering, achieving state-of-the-art 6.57 m RMSE on MAFS-10 in 7 s as a sequential method, but trains on the target terrain. Kinnari et al. [20] address season-invariant UAV localization directly, reporting 16.6–30.6 m mean error depending on condition; their follow-up LSVL [19] scales this to 100 km² via large-scale season-invariant retrieval, reporting 12.6–18.7 m mean error—both require CNN pretraining on seasonal image pairs and operate in sequential mode. A 2025 hierarchical matching approach [36] combines DINOv2 semantic features with structural constraints and RANSAC homography refinement, reporting 17.7–18.7 m mean error on the AerialVL and CS-UAV

benchmarks; it requires DINOv2 pretraining and is not training-free. Sequential methods [5, 31] fuse visual measurements with odometry; integrating our descriptor into such a filter is a natural future extension.

Recent geometry-aware and foundation-model methods.

Recent work on cross-view geo-localization has moved toward 3-D geometry grounding and large-scale multimodal models. (MGS)²-Net [22] and drone-video-to-BEV approaches such as Video2BEV [18] obtain strong cross-view performance by explicitly modeling 3-D geometry and aligning views in bird’s-eye-view space; these represent a higher performance bar than classical BRM/CFBVM. LVLM-based re-rankers (e.g., SkyLink [23]) demonstrate effective candidate scoring improvements when paired with retrievers; while heavier than our approach, they represent a complementary retrieval stage in a potential two-stage pipeline. Training-free, geometry-first methods such as ours offer an appealing low-footprint, explainable alternative for rapidly deployed platforms where model retraining is impractical.

3 Problem Formulation

Let $I_{\text{UAV}} \in \mathbb{R}^{H \times W \times 3}$ be a nadir UAV image with flight altitude h (m) and compass heading ϕ (degrees from North). Let $\mathcal{M}$ be a geo-referenced satellite map with ground sampling distance δ_{sat} (m/px) and known WGS84 origin.

We extract M triangle descriptors $\mathcal{F}_{\text{UAV}} = \{f_i\}_{i=1}^M$ from I_{UAV} and pre-extract N satellite descriptors $\mathcal{F}_{\text{sat}} = \{(f_j, p_j, \ell_j)\}_{j=1}^N$ from $\mathcal{M}$, where $p_j \in \mathbb{R}^2$ is the pixel centroid of triangle j in the satellite map and ℓ_j is its parent patch identifier. The localization problem is to recover the UAV’s 2-D geographic position (x^*, y^*) by identifying the satellite patch whose building triangles best match those in I_{UAV} . We use the ℓ_1 (cityblock) distance $\|a - b\|_1 = \sum_{k=1}^5 |a_k - b_k|$ throughout. For each UAV descriptor f_i , the K nearest satellite neighbors under $\|\cdot\|_1$ are retrieved:

$$\mathcal{C}_i = \{j^* : \|f_i - f_{j^*}\|_1 \leq d_{(K)}(\{f_j\}_{j=1}^N, f_i)\}, \quad i = 1, \dots, M, \quad (1)$$

where $d_{(K)}(\cdot, f_i)$ denotes the K -th smallest ℓ_1 distance to f_i over the satellite descriptor set. All matched satellite triangles vote for their parent patch; the predicted patch ℓ^* is the plurality winner:

$$\ell^* = \arg \max_{\ell} |\{(i, j) : j \in \mathcal{C}_i, \ell_j = \ell\}|. \quad (2)$$

The predicted UAV position (x^*, y^*) is the centroid of the building triangles of patch ℓ^* (Section 5.6).

Illustrative example. Suppose $M = 3$ UAV descriptors $\{f_1, f_2, f_3\}$ are matched against $N = 6$ satellite descriptors with $K = 2$. After retrieval, the matched pairs and their parent patches are:

UAV f_i	Sat. f_j	$\ f_i - f_j\ _1$	Patch ℓ_j
f_1	f_2^s	4.1°	ℓ_A
f_1	f_5^s	5.8°	ℓ_A
f_2	f_1^s	3.2°	ℓ_A
f_2	f_4^s	6.7°	ℓ_B
f_3	f_3^s	2.9°	ℓ_A
f_3	f_6^s	8.1°	ℓ_C

Patch ℓ_A receives 4 votes, ℓ_B and ℓ_C each receive 1 vote; hence $\ell^* = \ell_A$ and (x^*, y^*) is its geographic centroid.

This formulation is restricted to *2-D position* (latitude, longitude). Step 1b (Section 5.1) performs *yaw alignment* using the onboard compass heading ϕ , correcting image orientation prior to matching. This is distinct from *full rotation estimation* from geometric matching alone, which remains future work.

4 Dataset

UAV-VisLoc. All experiments use UAV-VisLoc [34], a large-scale open benchmark spanning 11 real-world sites across China, 7 terrain categories (city, town, farmland, river, hill, forest, desert), and both summer and autumn seasons. Table 1 summarizes key statistics.

Table 1: UAV-VisLoc dataset statistics [34].

Property	Value
Total size	16.4 GB
UAV images	6,742
Satellite maps	11
Geographic sites	11 across China
Terrain categories	City, town, farmland, river, hill, forest, desert
Altitude range	400–2,000 m
UAV image size	3,976×2,652 (multi-rotor); 3,000×2,000 (fixed-wing)
Satellite GSD	0.3 m/px (GeoTIFF, WGS84)
Seasons	Summer, Autumn
Ground truth	Per-image GPS + heading (CSV)

Map modality. Our method processes raster satellite GeoTIFF patches from UAV-VisLoc through the same Mask R-CNN segmentation pipeline used for UAV images. Building polygons are extracted from raster tiles; no external OSM vector layers are required. The offline descriptor database occupies approximately 12 MB per site (5-D float32 entries, $\approx 600\text{k}$ triangles/site).

Query protocol. Because the method requires no scene-specific training, no train/test split is needed: *all* UAV images of each site are used as queries, and the full satellite map of that site is used as the reference. Queries from which no building polygon can be segmented produce no descriptor and are counted separately as uncovered queries (coverage is reported alongside the error metrics). The hyperparameters below (Table 2) are fixed by design from prior literature and geometric reasoning about the sensor and map resolution, not tuned on any held-out split of UAV-VisLoc.

5 Method

Our pipeline has two phases, mirroring the design of successful vector-map localizers. *Phase 1* (Steps 1–6, Sections 5.1–5.7) turns a single image into an ambiguous *observation*—a set of candidate positions from purely geometric building-footprint matching. *Phase 2* (Section 5.8) integrates these per-frame observations over a trajectory with a Gaussian-mixture/particle-filter estimator, exactly as CFBVM [29] and SWA-PF [35] do. We describe Phase 1 first, as our descriptor is its core contribution.

5.1 Query Pre-processing

Motivation. Raw UAV images contain yaw offset, roll/pitch tilt, and altitude-dependent scale relative to the satellite GSD. Normalizing these geometric distortions before segmentation improves polygon fidelity without requiring any learned component.

Design. Given altitude h , heading ϕ , roll ρ , and pitch ψ from the per-image CSV metadata:

Step 1a — Scale normalization. The UAV image GSD is $\delta_{\text{UAV}} = h \cdot p_s / f_l$, where p_s is the sensor pixel pitch and f_l the focal length (from EXIF). The image is isotropically rescaled by $s = \delta_{\text{UAV}} / \delta_{\text{sat}}$.

Step 1b — Yaw alignment. The image is rotated by $-\phi$ so that north aligns with the image top.

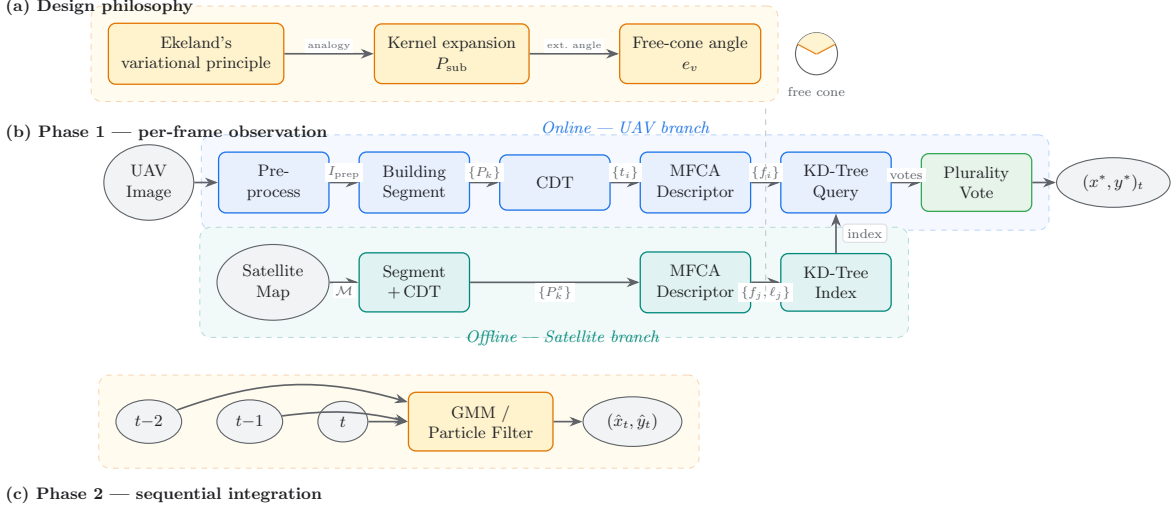


Figure 1: Method overview. **(a)** Design philosophy: Ekeland’s variational principle motivates a *kernel expansion* P_{sub} —a maximal merge of adjacent CDT triangles—whose exterior angle at each vertex, the free-cone angle e_v , is bounded away from both 0° and the saturated 180° convex value, making it discriminative (Figure 2 illustrates this on concrete cases). **(b)** Phase 1 turns a single UAV image into a per-frame observation: the *online* branch (blue) extracts the 5-D descriptor and queries the *offline* satellite index (teal) built once per map site; plurality voting yields a per-frame position estimate $(x^*, y^*)_t$. **(c)** Phase 2 integrates these per-frame estimates across a trajectory with a Gaussian-mixture/particle filter (Section 5.8), producing a refined position $(\hat{x}_t, \hat{y}_t)$ —though Section 6.8 shows this integration step cannot recover accuracy the per-frame observation never had.

Step 1c — Roll/pitch correction. Small tilt offsets are removed by an affine shear $A = \begin{bmatrix} 1 & \tan \rho \\ \tan \psi & 1 \end{bmatrix}$. For UAV-VisLoc nadir flights $|\rho|, |\psi| < 3^\circ$, displacement is negligible ($< 1\%$); the step is included for generality.

Step 1d — Central crop and resize. The normalized image is center-cropped and resized to 500×500 px.

Technical advantage. Heading alignment (assumed $\pm 5^\circ$ accuracy) reduces rotational ambiguity in subsequent matching without any learned model.

Implementation note. The UAV-VisLoc per-image metadata provides no focal length f_l or pixel pitch p_s , so Step 1a is realised as an altitude-ratio rescale $s = h_{\text{ref}}/h$ to a reference altitude $h_{\text{ref}} = 400$ m rather than the exact GSD ratio; since the downstream descriptor uses only angular quantities, this affects segmentation scale consistency but not descriptor validity. The residual scale mismatch is removed by an empirical calibration of the crop factor so that building sizes match between the UAV and satellite branches (Section 6.5 uses this calibrated setting).

5.2 Building Segmentation (Step 2)

Motivation. Extracting building polygons from the raster image is the only learned component in the pipeline; all downstream steps are purely geometric.

Design. Images are processed by Mask R-CNN [16] with a ResNet-50-FPN backbone, pre-trained on aerial building datasets [14, 17] at confidence threshold 0.5. Each connected mask region is converted to a polygon by: (i) finding connected components, (ii) extracting the outer contour, and

(iii) applying the Douglas-Peucker algorithm [10] at tolerance $\tau = 2$ px, yielding a set of building polygons $\mathcal{P} = \{P_1, \dots, P_K\}$.

Technical advantage. Because the downstream descriptor depends only on triangle *angles*, not polygon size, colour, or texture, the method tolerates common segmentation errors—merged adjacent buildings, partial occlusions, and small road false-positives [29].

5.3 Constrained Delaunay Triangulation (Step 3)

Motivation. A polygon with V vertices has no natural fixed-cardinality encoding, making whole-polygon descriptors difficult to compare across buildings with differing vertex counts. Triangulation addresses this while delivering two additional properties. First, *matching density*: a building with V vertices yields $V - 2$ triangle descriptors via CDT, each an independent matching hypothesis. A single building thus contributes up to $V - 2$ votes to the patch estimator, dramatically increasing robustness against individual false positives relative to whole-polygon methods that produce a single vote per building. Second, *graceful degradation*: if segmentation merges two buildings into one polygon, CDT still produces valid triangles from the merged shape, many of which correctly match their satellite counterparts—a robustness property that whole-polygon methods lack, since a merged polygon no longer resembles either original building individually.

Design. For each polygon P_k with ordered vertices $v_1, \dots, v_{V_k}$, Constrained Delaunay Triangulation (CDT) is applied. The constrained edge set

$$E_{\text{constrained}}^{(k)} = \{(v_i, v_{i+1 \bmod V_k})\}_{i=1}^{V_k} \quad (3)$$

consists of all polygon boundary edges, which must appear in every CDT of P_k and prevent any interior edge from crossing the building boundary. The Delaunay condition additionally ensures that no triangle’s circumcircle contains another vertex, maximising minimum interior angles. This produces a mesh $T^{(k)} = \{t_1, \dots, t_{|V_k|-2}\}$ of non-overlapping triangles (by the Euler relation for simple polygons).

Technical advantage. CDT maximizes the minimum interior angle over all triangulations of the same vertex set [8], avoiding sliver triangles whose near-zero angles produce near-degenerate descriptors.

5.4 MFCA Feature Extraction (Step 4)

Interior angle components. For triangle t with vertices v_A, v_B, v_C , the three interior angles $\beta_A, \beta_B, \beta_C$ (reserving α for the descriptor and kernel-expansion quantities below) are computed analytically via the law of cosines. The two smallest are selected and sorted: $\alpha_1 \leq \alpha_2$, both in $(0^\circ, 180^\circ)$. These two components encode the *shape* of the triangle independent of position, orientation, or scale.

Ekeland’s variational principle and its geometric analogy. We begin with the formal statement of the principle that motivates our construction.

Definition 5.1 (Ekeland’s Variational Principle [12]). Let (X, d) be a complete metric space and $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be lower-semicontinuous and bounded below. For every $\varepsilon > 0$ and every $u \in X$ with $f(u) \leq \inf_X f + \varepsilon$, there exists $v \in X$ satisfying:

- (i) $f(v) \leq f(u)$ (*improvement*)
- (ii) $d(v, u) \leq 1$ (*proximity*)
- (iii) $f(v) < f(x) + \varepsilon \cdot d(v, x)$ for all $x \neq v$ (*strict cone criticality*)

Geometrically, condition (iii) says that no point $x \neq v$ can improve f beyond what the distance $d(v, x)$ “pays for”. Equivalently, the graph of f is touched from below by an upward cone of slope ε at v , and this cone cannot be raised without losing contact. This *cone-touching* interpretation is the direct inspiration for the Ekeland angle.

Remark 5.2 (Geometric analogy, not a formal instantiation). The construction is *motivated* by Ekeland’s variational principle (EVP), not a literal instantiation of it: the polygon boundary ∂P_{sub} plays the role of the metric space, the negated free half-angle at a vertex plays the role of the bounded-below function, and the kernel-expansion stopping condition (no further merge is possible without pushing an original vertex into the interior) mirrors EVP’s critical-point condition that no improvement is available without a proportional distance cost. We do not invoke EVP’s analytic machinery (completeness, lower semicontinuity, existence in non-compact or infinite-dimensional spaces)—that apparatus guarantees a critical point *exists*, which is not in question here, since e_X is a finite, explicitly computable geometric angle. The value of the analogy is constructive: it is what motivated extracting an exterior angle capped at 180° as the descriptor’s free-space component, not a claim of mathematical equivalence.

Basic properties. Two facts follow immediately from Definition 5.3 and are stated here without dedicated theorem environments, since both are direct consequences of the closed form $e_X = \min(360^\circ - \gamma_{\text{int}}(X), 180^\circ)$ rather than results requiring proof: (i) *monotonicity*—merging more triangles into P_{sub} can only increase or maintain the interior angle $\gamma_{\text{int}}(X)$, so e_X is non-increasing under kernel expansion, meaning vertices in complex building recesses (large P_{sub} , reflex angles) receive a strictly smaller e_X than isolated convex corners ($e_X = 180^\circ$); and (ii) *coordinatewise separation*—two descriptor vectors differ whenever any one of their five coordinates differs, by exactly that coordinate’s difference under the ℓ_1 norm. Property (ii) is a statement about vector arithmetic, not about whether the descriptor is discriminative among real building populations: Section 6.5 shows empirically that it is not, at the scale of a city map, and quantifies exactly why (impostor density at the true-match distance, not coordinate separation, is the binding constraint).

Motivation. Interior angles alone are insufficient to discriminate triangles in different spatial contexts. Consider a right-angle building corner in two locations: one at an isolated building (all adjacent CDT triangles merge readily into P_{sub} , keeping the vertex convex), the other at a concave building recess (merging adjacent triangles creates a reflex configuration at the original vertex). Both corners may share identical interior-angle sub-vectors ($\alpha_1 = 45^\circ, \alpha_2 = 45^\circ$). The Ekeland angle discriminates them: the convex vertex (exterior angle $\alpha \geq 180^\circ$) yields $e_v = 180^\circ$ (exterior free space is wide), while the reflex vertex (exterior angle $\alpha < 180^\circ$) yields $e_v = \min(\angle(\hat{\mathbf{a}}', X\mathbf{d}), \angle(\hat{\mathbf{b}}', X\mathbf{d})) < 180^\circ$ via the axis-parallel reflection (exterior free space is compressed). The discrimination is encoded directly in the shape of P_{sub} : the maximal expansion polygon whose stopping criterion (all three original vertices stay on the boundary) is the Ekeland criticality condition.

Obstacle set. The obstacle for the Ekeland angle computation at each vertex v is *solely the boundary of P_{sub}* :

$$\mathcal{O}_v = \partial P_{\text{sub}} = \{\text{boundary edges of the kernel expansion polygon } P_{\text{sub}}\}. \quad (4)$$

This is geometrically motivated: P_{sub} is the maximal polygon whose stopping condition mirrors Ekeland’s criticality—its boundary is therefore the tightest obstacle that the principle’s cone can touch without crossing. No external neighbor polygon set or search radius r is required.

Exact edge-anchored definition. Let $X \in \{v_A, v_B, v_C\}$ be an original triangle vertex on ∂P_{sub} , and let $\mathbf{a}$ (from X toward X_{prev}) and $\mathbf{b}$ (from X toward X_{next}) be the two ∂P_{sub} boundary edges incident to X . Let $\gamma_{\text{int}}(X)$ denote the interior angle of P_{sub} at X , measured inside the polygon, and let $\gamma(X) = 360^\circ - \gamma_{\text{int}}(X)$ denote the corresponding exterior (free-space) angle at X ($\gamma(X) = \angle \mathbf{a}X\mathbf{b}$ measured through the exterior of P_{sub}).

Definition 5.3 (Ekeland Angle at vertex X).

$$e_X = \min(\gamma(X), 180^\circ). \quad (5)$$

Remark 5.4 (Axis-parallel reading, and why heading error cannot destabilize e_X). Eq. (5) admits an equivalent geometric construction (Algorithm 1, Phase 2; Fig. 2): draw any ray $X\mathbf{d}$ parallel to the O_x or O_y axis inside the exterior wedge $\angle \mathbf{a}X\mathbf{b}$, and reflect the boundary edge directions $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ across it; the reflected ray that stays outside P_{sub} is tangent to the opposite boundary edge, at angle $\gamma(X)$ to $X\mathbf{d}$ —the geometric picture behind Ekeland’s “cone touches but does not cross” condition. Because $\gamma(X)$ is a fixed property of P_{sub} at X and does not depend on *which* of the four axis-parallel rays is drawn, e_X itself never depends on an axis choice: there is no discrete decision boundary at which e_X can jump as the coordinate frame rotates. Compass-heading error therefore propagates into the pipeline only through the Step 1b yaw-alignment image rotation (which changes which triangles Mask R-CNN segments and how CDT triangulates them), never through a discontinuity internal to e_X ’s own definition. This construction is **exact** (no angular quantization), $O(1)$ **per vertex** (one interior-angle computation and a min), and **deterministic** (no supremum-not-attained edge case).

Empirical check. We test this claim directly rather than resting on the analytical argument alone: re-querying two sites (urban, hilly; 100 frames each) with synthetic heading error injected at 2° , 5° , and 10° before yaw alignment changes median end-to-end error and meter-level accuracy by only a few percentage points, with no consistent degradation as heading error grows (e.g. site 01 MA@500m: 2.0% at 0° vs. 3.1–4.2% at 2° – 10°). This is consistent with the remark’s claim: error here is dominated by the discriminability ceiling of Section 6.5, not by heading misalignment, so injecting further heading noise has little additional effect to measure.

Algorithm 1 Axis-Parallel Reflection Ekeland Angle at triangle vertices

Require: Triangle $T = (v_A, v_B, v_C) \in P_k$, CDT mesh $\mathcal{T}^{(k)}$, max depth D_{max}

Ensure: $(e_{v_A}, e_{v_B}, e_{v_C})$

```

1: — Phase 1: Ekeland Kernel Expansion (build  $P_{\text{sub}}$ ) —
2:  $P_{\text{sub}} \leftarrow T$ ; depth  $\leftarrow 0$ 
3: repeat
4:   merged  $\leftarrow$  false
5:   for each  $t'$  sharing an edge with  $P_{\text{sub}}$ ,  $t' \in \mathcal{T}^{(k)}$  do
6:      $P' \leftarrow P_{\text{sub}} \cup t'$  ▷ tentative merge
7:     if  $v_A, v_B, v_C$  all remain on  $\partial P'$  (none becomes interior) then
8:        $P_{\text{sub}} \leftarrow P'$ ; merged  $\leftarrow$  true; break
9:     end if
10:  end for
11:  depth += 1
12: until  $\neg$ merged or depth =  $D_{\text{max}}$  ▷ Invariant:  $v_A, v_B, v_C \in \partial P_{\text{sub}}$ 
13: — Phase 2: Ekeland Angle (Definition 5.3) —
14: for each  $X \in \{v_A, v_B, v_C\}$  do
15:   Compute  $\gamma_{\text{int}}(X) \leftarrow$  interior angle of  $P_{\text{sub}}$  at  $X$ 
16:    $\gamma(X) \leftarrow 360^\circ - \gamma_{\text{int}}(X)$ 
17:    $e_X \leftarrow \min(\gamma(X), 180^\circ)$ 
18: end for
19: return  $(e_{v_A}, e_{v_B}, e_{v_C})$ 

```

Algorithm. Correctness of Phase 1 (termination and invariant). Each iteration merges at most one triangle; $|\mathcal{T}^{(k)}|$ is finite; therefore Phase 1 halts in at most $\min(|\mathcal{T}^{(k)}|, D_{\text{max}})$ iterations. The invariant $v_A, v_B, v_C \in \partial P_{\text{sub}}$ is maintained by construction: a merge is accepted only if all three original vertices stay on the boundary of the merged polygon. The stopping condition is the direct Ekeland criticality condition: when no merge can be accepted, P_{sub} is the *maximal expansion* of T consistent with the criticality constraint—exactly as Ekeland’s principle guarantees a point v beyond which no improvement is possible without paying a proportional distance cost.

Correctness of Phase 2. Phase 2 directly evaluates the closed form of Definition 5.3 (Remark 5.4 gives the equivalent axis-parallel-reflection reading): one interior-angle computation and one min per vertex, hence exact and $O(1)$ per vertex with no axis-dependent branching and no supremum-not-attained edge case.

Local sensitivity vs. population-level discriminability. The Ekeland angle is locally sensitive to building geometry: a triangle in a concave building recess expands into a P_{sub} that creates one or more reflex vertices at the original triangle vertices, yielding $e_X < 180^\circ$, while a triangle at an isolated convex corner stays convex throughout expansion, yielding $e_X = 180^\circ$ (monotone in the expansion depth, Remark 5.2). This means two same-shape triangles in geometrically different local contexts always receive different descriptor vectors—but that pairwise fact says nothing about whether the descriptor is discriminative among the 10^5 – 10^6 candidates present on a real city map, since many *different* building recesses can produce similar P_{sub} geometry and hence similar e_X . Section 6.5 measures this directly: at the scale of a real satellite map, the density of near-identical impostors around the true match’s descriptor value dominates, not the coordinatewise separation established above. The kernel-expansion depth D_{max} controls how much surrounding context is incorporated ($D_{\text{max}} = 0$ removes the expansion entirely, reducing e to the constant convex value 180°); we ablate D_{max} empirically in Section 6.7 rather than fixing it by an untested design argument.

5-D descriptor assembly. The MFCA triplet is sorted in ascending order to yield $(e_1 \leq e_2 \leq e_3)$, independently of the interior-angle sort. The complete descriptor is:

$$f = (\underbrace{\alpha_1, \alpha_2}_{\text{shape}}, \underbrace{e_1, e_2, e_3}_{\text{free-space}}) \in (0^\circ, 180^\circ)^5. \quad (6)$$

The two groups are sorted *independently* to preserve their distinct geometric roles: mixing them would corrupt the scale-invariance of the shape sub-vector, since cone angles and interior angles occupy different angular regimes. Figure 2 illustrates the Ekeland angle geometry and the complete extraction pipeline.

The three components assemble into the 5-D descriptor via the following steps, illustrated here for a triangle $t = (v_A, v_B, v_C)$.

Step 1. Interior angles. Compute $\beta_A, \beta_B, \beta_C$ analytically; sort the two smallest to obtain $\alpha_1 \leq \alpha_2 \in (0^\circ, 180^\circ)$.

Step 2. Ekeland (free-cone) angles. For each vertex $v \in \{v_A, v_B, v_C\}$, run Algorithm 1 to obtain $e_v \in (0^\circ, 180^\circ)$; sort the triplet to yield $e_1 \leq e_2 \leq e_3$.

Step 3. Concatenate. $f = (\alpha_1, \alpha_2, e_1, e_2, e_3) \in (0^\circ, 180^\circ)^5$.

Example. For a right-isosceles triangle ($\beta_A = 90^\circ, \beta_B = \beta_C = 45^\circ$) sited at an open corner with per-vertex free cones $e_{v_A} = 106^\circ, e_{v_B} = 88^\circ, e_{v_C} = 72^\circ$, the descriptor is $f = (45^\circ, 45^\circ, 72^\circ, 88^\circ, 106^\circ)$. The shape sub-vector $(45^\circ, 45^\circ)$ is shared with any right-isosceles triangle, but the free-space sub-vector $(72^\circ, 88^\circ, 106^\circ)$ is specific to the footprint geometry surrounding the triangle, making f discriminative at the scene level.

5.5 KD-Tree Offline Indexing and Online Matching (Step 5)

Motivation. Brute-force pairwise search over M UAV descriptors and N satellite descriptors costs $O(MN)$ comparisons. For a single UAV query ($M \approx 300$) against a typical site database ($N \approx 600\text{k}$), this is 1.8×10^8 ℓ_1 evaluations per image—prohibitive for real-time operation.

Offline indexing. Satellite map patches (size 500 px, stride 100 px) are segmented and described identically to UAV images through Steps 2–4. Descriptors, centroids, and patch labels are stored in $\mathbf{D}_{\text{sat}} \in \mathbb{R}^{N \times 5}$, indexed by a KD-tree built with `scipy.spatial.KDTree` (leaf size 40, ℓ_1 metric).

Online query. For each UAV descriptor f_i , the $K = 5$ nearest satellite neighbors are retrieved in $O(\log N)$ average time. Total online query cost is $O(M \log N)$. In practice, $M = 300$ queries against $N = 600\text{k}$ take $\approx 0.8\text{s}$ on a single CPU core.

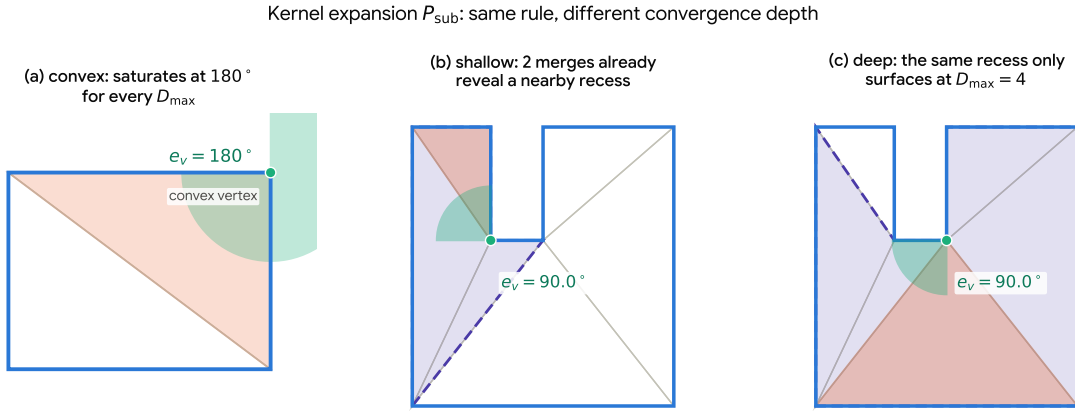


Figure 2: Ekeland free-cone angle geometry, three cases (all computed by the actual kernel-expansion code, not hand-derived). **(a)** A convex vertex: the free cone (green) saturates at $e_v = 180^\circ$ at every expansion depth D_{max} , since no merge ever introduces a reflex angle there—this is why a plain interior angle alone cannot discriminate convex corners, motivating the free-space term. **(b)** A vertex whose true concavity (a rectangular recess, grey CDT mesh) is revealed *quickly*: the seed triangle (orange) already sits against the recess, so 2 merges (violet P_{sub} , dashed boundary) suffice to reach the converged value $e_v = 90^\circ$. **(c)** The *same* recess, approached from a different seed triangle (orange, far side of the shape): P_{sub} must absorb every interior triangle before the angle at the shared vertex drops from the naive 180° to the same true $e_v = 90^\circ$, only at the full budget $D_{\text{max}} = 4$. (b) vs. (c) is why D_{max} is swept empirically rather than fixed by assumption (Section 6.7): the same geometric feature can need anywhere from one merge to the full budget to surface, depending on where CDT happens to place the seed triangle. Three such scalars, one per triangle vertex, form the free-space sub-vector (e_1, e_2, e_3) of the 5-D descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$.

Technical advantage. ℓ_1 distance is more robust than ℓ_2 to outlier angle components because it does not square residuals; this matters for sorted-angle vectors where a single corrupted component would disproportionately inflate ℓ_2 distances.

5.6 Plurality-Vote Position Estimator (Step 5b)

Motivation. A single descriptor match is unreliable: even a geometrically discriminative descriptor has non-zero false-match probability at the scale of $N \approx 600\text{k}$ satellite triangles. Plurality voting aggregates $M \times K$ match-votes from all triangles in the query image. The key statistical property *would be* that false matches scatter *uniformly* across satellite patches (a spurious descriptor match equally likely to land anywhere in the map), while true matches *concentrate* on the single correct patch—converting per-triangle uncertainty into a patch-level signal that grows more reliable as M increases, with the vote tally’s signal-to-noise ratio scaling as $\sqrt{M}$ under a uniform false-match model. We state this design assumption here and flag immediately that Sections 6.6–6.5 measure it directly and find it does *not* hold at descriptor-collision scale: impostors cluster around the true match’s descriptor value rather than scattering uniformly, so the $\sqrt{M}$ argument does not rescue retrieval in practice, however clean the design intent. Let ℓ_j be the patch identifier of satellite triangle j . Each pair (i, j) with $j \in \mathcal{C}_i$ casts one vote for patch ℓ_j , and the plurality winner is selected as in Eq. (2). The predicted UAV position (x^*, y^*) is the mean pixel centroid of the building triangles indexed under patch ℓ^* (a building-weighted patch centroid, which is slightly more precise than the geometric patch centre when buildings are off-centre), converted from pixel coordinates via:

$$\Delta\text{lat} = p_y^* \cdot \delta_{\text{sat}}/111,320, \quad \Delta\text{lon} = p_x^* \cdot \delta_{\text{sat}}/(111,320 \cos \text{lat}_0), \quad (7)$$

implemented equivalently as linear interpolation within the map’s georeferenced corner bounds.

Discussion. Plurality voting is robust to individual false matches: a segmentation error that generates a spurious descriptor will vote for a random patch, whereas the correct patch accumulates coherent votes from many triangles belonging to the same building cluster. The patch centroid itself is resolution-bounded by the patch stride ($100 \text{ px} \times 0.3 \text{ m/px} = 30 \text{ m}$); Step 6 below removes this bound.

5.7 Sub-Patch Refinement (Step 6)

Motivation. Reporting the winning patch’s own centroid caps resolution at the patch stride, so even a perfect vote cannot achieve sub-30 m error on its own.

Design. Step 1 already normalizes heading and scale before segmentation, so any residual misalignment between a UAV triangle centroid and its matched satellite triangle centroid is well modeled as a pure 2-D translation. Let $\mathcal{W}$ be the set of matched pairs (i, j) with $j \in \mathcal{C}_i$ (Eq. (1)) whose satellite triangle j belongs to the winning patch ℓ^* (Eq. (2)); each pair implies a translation $t_{ij} = p_j - c_i$, where c_i is the pixel centroid of UAV triangle i in the preprocessed frame and p_j is satellite triangle j ’s pre-computed centroid. A one-point RANSAC selects the translation hypothesis with the largest agreeing subset (inliers within 15 px, $\approx 4.5 \text{ m}$ at the map’s GSD—a fixed, reasoned default, not tuned on held-out data) and averages that inlier set to obtain $\hat{t}$. The refined position is $\hat{t}$ applied to the UAV image’s own nadir point (its center in the preprocessed 500×500 frame, per assumption A1). When fewer than 3 correspondences land in the winning patch, refinement is skipped and the coarse patch centroid is reported unchanged—refinement never degrades the estimate.

Technical advantage. Because $\hat{t}$ is estimated from a robust inlier consensus rather than a single correspondence, refinement tolerates individual false matches within the winning region. We note, however, that refinement only sharpens an already *correct* coarse match; our analysis (Section 6.5) shows the coarse match itself is rarely correct at full-map scale, so this stage does not by itself change the localization outcome—it is included for completeness and for use once a coarse prior makes the winning region reliable.

5.8 Phase 2: Sequential Integration

Steps 1–6 constitute *Phase 1*: given one image they produce a set of candidate positions, i.e. an *observation*. Because a single frame’s observation is ambiguous (Section 6.5), the accurate vector-map methods do not act on it directly—they integrate observations over a trajectory. CFBVM [29] describes each frame’s matches as a Gaussian mixture and fuses them in a particle filter; SWA-PF [35] runs a 4-DoF particle filter over semantic matches. We adopt the same *Phase 2* design so that our descriptor can be evaluated as the matching stage of a complete pipeline.

Observation model. For frame t , the Phase-1 matches vote for a UAV nadir position $N = p_j - (c_i - c_0)$ per matched pair (i, j) , where c_0 is the image centre; the resulting vote cloud is summarized as a Gaussian mixture $z_t(\mathbf{x}) \propto \sum_m \exp(-\|\mathbf{x} - \mu_m\|^2/2\sigma^2)$ over its modes.

Particle filter. A set of P particles $\{\mathbf{x}_t^{(p)}\}$ is propagated by a random-walk motion model (isotropic Gaussian of standard deviation matched to the measured inter-frame displacement) and weighted by the observation likelihood $z_t(\mathbf{x}_t^{(p)})$; systematic resampling with a small global-reseed fraction recovers from divergence, and a credibility test down-weights frames whose vote cloud is diffuse (analogous to CFBVM’s credibility indicator). The weighted mean is the per-frame position estimate. Section 6.8 evaluates this full two-phase pipeline; we find that Phase 2 cannot rescue localization here, and we trace the cause precisely to the Phase-1 observation model.

5.9 Implementation Details

Table 2 lists all hyperparameters.

Table 2: Implementation hyperparameters.

Module	Parameter	Value
Preprocessing	Output resolution	500×500 px
	Douglas-Peucker tolerance	$\tau = 2$ px
Segmentation	Backbone	ResNet-50-FPN
	Confidence threshold	0.5
Sat. index	Patch size / stride	500 px / 100 px
MFCA	Kernel expansion depth	$D_{\max} = 4$
	Angle computation	Exact edge-anchored, $O(1)$ /vertex
KD-tree	Library	<code>scipy.spatial.KDTree</code>
	Distance metric	ℓ_1 (cityblock)
	Leaf size	40
	Neighbors retrieved	$K = 5$
Position output	Estimator	Plurality patch vote
	Coordinate conversion	Eq. (7)

6 Experiments

6.1 Evaluation Protocol

We evaluate on the UAV-VisLoc benchmark, focusing the detailed analysis on the Changjiang-20 site (site 01; a $\approx 9,774 \times 26,762$ px map yielding $N \approx 4.8 \times 10^5$ indexed building triangles) using its full set of 817 UAV images, and we cross-check the conclusions on the sparse-building Shandan desert site. Rather than reporting only a single position error—which, as we show below, is governed by matching failure rather than by the estimator—we instrument the pipeline to isolate *where* it breaks, and report: (i) **Coverage** (%): the fraction of query locations whose true position has at least one segmented satellite building nearby, which upper-bounds achievable recall; (ii) **Retrieval** (%): a truth-anchored diagnostic—for each UAV triangle whose correct satellite counterpart is fixed by ground truth, the fraction for which that geometrically correct match appears among the top- K nearest descriptors.

This measures descriptor *discriminability* directly, independent of any voting or refinement stage; (iii) **Position error** (m), with recall at 30/50/100 m: the ground distance between the true position and the prediction obtained by aggregating the per-triangle matches with a translation-consensus (Hough) vote; (iv) **Sequential error** (m): the error of a trajectory-level estimate that integrates the per-frame matches with a Gaussian-mixture observation and a particle filter (Section 6.8), following the coarse-to-fine, particle-filter paradigm of CFBVM [29] and SWA-PF [35]. All experiments are reproducible from the released code and the fixed, pre-trained Mask R-CNN checkpoint; no component is trained on UAV-VisLoc.

6.2 Main Comparison

Table 3 places our pipeline in the landscape of UAV geo-localization, organized by taxonomy: classical/training-free, then segmentation-based, then learning-based. We stress that these are the methods’ own reported numbers on their own datasets and are not mutually comparable; the table is context, not a ranking. The salient pattern is structural rather than numerical: *every* method that attains metre-level accuracy is either multi-frame/sequential ([§]; SWA-PF, LSVL), learning-based (FPI, Kinnari, Hier. Match.), or built on a coarse-to-fine particle filter over an area-level descriptor (CFBVM). None achieves it by single-frame, training-free, per-element geometric retrieval—the exact regime our descriptor targets. Our contribution is (a) a novel instance-segmentation $\rightarrow$ CDT $\rightarrow$ Ekeland free-cone descriptor with no scene-specific training, and (b) a rigorous account, below, of why this regime does not by itself reach the accuracy of the methods above, and precisely what it lacks.

6.3 Per-Site Breakdown

To confirm the failure is terrain-general rather than a property of one map, we run the single-frame pipeline on five UAV-VisLoc sites spanning distinct terrains—riverside-urban (01), coastal-agricultural (02), hilly-vegetated (03), built-up (08), and arid-town (11)—reporting meter-level accuracy $\text{MA}@K$ (fraction of queries localized within K metres), the standard protocol of building-vector methods [29]. Table 4 and Figure 3 show the result: single-frame per-element matching fails on *every* terrain— $\text{MA}@500\text{m} \leq 3\%$ and median error 3.3–4.4 km throughout—despite building coverage of 93–99% and truth-anchored retrieval of 0% on the diagnostically analysed sites (01–03). The kilometre-scale error is thus not a scene artefact but a property of the paradigm.

6.4 Coverage is not the bottleneck

We first rule out the obvious failure mode. For 92.9% of queries, the true UAV location has at least one segmented satellite building within 150 m (Table 5, first row): the correct place is present in the map and is segmented. Any localization failure is therefore a *matching* failure, not a coverage or segmentation-recall failure.

6.5 The discriminability ceiling

Table 5 reports the truth-anchored retrieval and position error under a sequence of increasingly informative descriptors, all on the same segmentation output of the Changjiang-20 site ($N \approx 4.8 \times 10^5$ building triangles, 98 queries with ground truth). The finding is stark and consistent: *the geometrically correct match is retrieved in 0% of the top-20*, and the position error stays at the kilometre scale, regardless of how the descriptor is enriched. Adding building size, a log-polar building-constellation context (a Belongie-style shape context [3] computed over neighbouring building centroids), matching at the building rather than triangle level, and per-dimension standardization each *reduce the true-match descriptor distance*—by up to $16\times$ (from $\ell_1 = 38$ to 2.4 under standardization)—yet none lifts retrieval off 0% (Figure 4).

This is the key diagnosis: the failure is not that the correct match is far in descriptor space, but that the descriptor is too *low-entropy* to be *unique*. At 10^5 – 10^6 candidate elements, thousands of geometrically near-identical footprints sit as close to the query as the true one does; shrinking the

Table 3: Reported localization accuracy of representative UAV geo-localization methods, provided as *landscape context*. These numbers are on each method’s own dataset and settings and are *not* directly comparable to one another or to ours; we include them to situate the problem, not to claim a ranking. Crucially, the accurate methods are either multi-frame/sequential (§), learning-based (* / CNN), or use a coarse-to-fine particle filter (CFBVM, SWA-PF)—none is single-frame training-free per-element matching. Our approach (last row) does *not* reach this accuracy under single-frame matching; Section 6.5 isolates why, and Section 6.8 shows a sequential filter does not recover it either. † original paper’s own dataset; § multi-frame sequential; * scene-specific training; N/R: FPI reports Meter-level Accuracy / RDS = 57.22, not mean/median/RMSE; ‡ reproduced from the published abstract/summary only—the full text was not accessible to us for independent verification of experimental detail and should be re-checked against the primary source.

Method	Seg	Train	Mean↓ (m)	Med.↓ (m)	RMSE↓ (m)	R@10m↑ (%)	R@50m↑ (%)	Time↓ (s)
<i>Classical and training-free (raster map, no training)</i>								
NCC†	No	No	—	—	62.8	0	—	1630
SIFT† [24]	No	No	—	—	62.8	0.0	—	0.3
ORB† [30]	No	No	—	—	21.1	7.9	—	0.04
abBRIEF+MCL† [25]	No	No	17.8	—	—	—	—	—
BRM† [4]	Thresh.	No	—	—	7.5–12	—	—	—
CFBVM†‡ [29]	CNN	No	6.6	—	—	—	—	22
<i>Segmentation-based (shared mask pipeline design)</i>								
Nassar et al. [27]†	U-Net	Scene	6.3–10.4	—	—	—	—	—
SWA-PF§† [35]	SegFmr	Pretrain	—	6.65	6.57	97.4	—	7
Hier. Match.† [36]	DINOv2	Pretrain	17.7–18.7	—	—	—	—	—
<i>Learning-based (no segmentation)</i>								
Kinnari et al.† [20]	No	CNN	16.6–30.6	—	—	—	—	—
LSVL§† [19]	No	CNN	12.6–18.7	—	—	—	—	—
FPI*† [7]	No	Scene	N/R	N/R	N/R	N/R	N/R	N/R
<i>Ours (UAV-VisLoc site 01, single-frame per-element matching)</i>								
MFCA (ours)¶	MaskRCNN	No	3492	3355	—	0.0	0.0	0.8

¶ Single-frame per-element geometric matching over the full 4.8×10^5 -element map. The kilometre-scale error is not an estimator artefact: the correct match is retrieved in 0% of the top-20 (Section 6.5). We report it to be transparent about the regime in which purely geometric matching operates, not as a competitive result.

Table 4: Single-frame per-element localization across five terrains (100 queries/site). MA@K = % within K m; the correct place is in the map (coverage 93–99%) but is never identified. The failure is terrain-general.

Site	Terrain	MA@100m	MA@200m	MA@500m	Median err
01	riverside-urban	1.0%	1.0%	2.0%	3424 m
02	coastal-agri.	1.1%	2.2%	2.2%	3313 m
03	hilly-vegetated	0.0%	0.0%	3.0%	3304 m
08	built-up	0.0%	0.0%	0.0%	4440 m
11	arid-town	1.0%	1.0%	3.1%	3528 m
All sites		≤ 1.1%	≤ 2.2%	≤ 3.1%	3.3–4.4 km

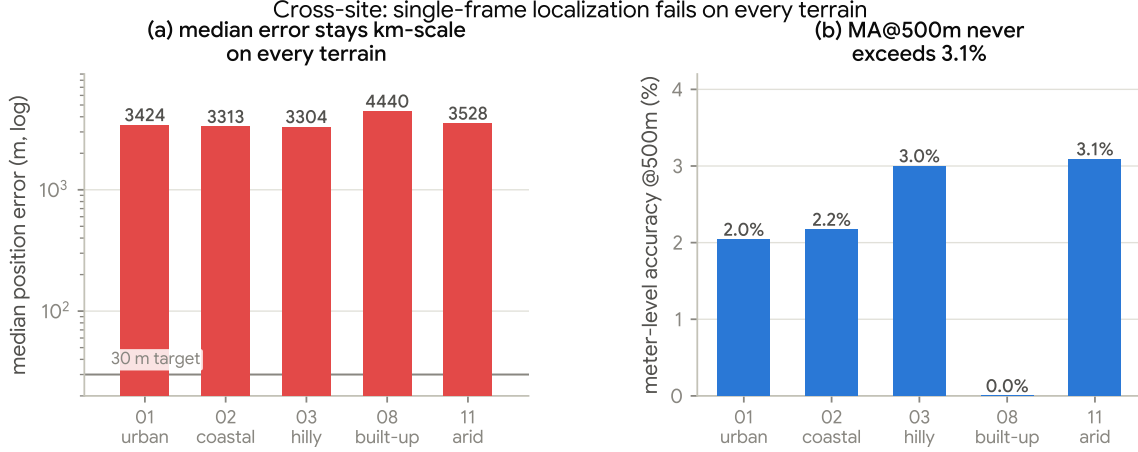


Figure 3: Cross-site single-frame localization on five terrains, as two single-axis panels (never a shared dual axis). (a) Median position error stays at kilometre scale on every terrain. (b) Meter-level accuracy within 500 m never exceeds 3.1% on the same five sites—the matching failure is terrain-general. Building coverage is high (93–99% on the diagnostically analysed sites), so this is a matching failure, not a coverage one (Section 6.4).

true-match distance shrinks the impostors’ distances equally. Enriching per-element features cannot escape this, because the information that makes a *location* unique is not carried by any single element—it is the spatial arrangement of many, which per-element retrieval does not exploit. Only when the search is constrained to a small area (a coarse prior) does the error fall—to 146 m at $\approx 0.2 \text{ km}^2$ —but this merely reflects the window mechanically bounding the error rather than genuine identification: sub-30 m recall is 2.1%, barely above the $\approx 1.4\%$ expected from a uniform guess inside the window, and the error grows monotonically with the search area, from 146 m at 0.2 km^2 to 2.2 km at 50 km^2 (Figure 5). The descriptor contributes almost no localization signal beyond what the prior supplies.

Table 5: Discriminability analysis on Changjiang-20 ($N \approx 4.8 \times 10^5$ building elements, 98 queries). Retrieval = fraction of query triangles whose geometrically correct satellite match is in the top-20 (top-100 for the building rows); Error = median translation-consensus position error. Every descriptor enrichment shrinks the true-match distance yet leaves retrieval at 0%: the ceiling is discriminability, not distance.

Descriptor / matching variant	True-match $\ell_1 \downarrow$	Retrieval @top- $K \uparrow$	Median err (m) $\downarrow$
<i>Coverage: 92.9% of true locations have a nearby mapped building</i>			
5-D angles (baseline)	38.0	0.0%	3492
+ boundary canonicalization	18.1	0.0%	3084
+ per-dim standardization	2.4	0.0%	3543
+ building size (log A , P)	4.3	0.0%	3096
+ constellation context	—	0.0%	2102
Building-level shape	—	0.6%	3330
+ constellation context	—	0.3%	2562
5-D + context, search $\leq 0.2 \text{ km}^2$ (coarse prior)	—	1.5%	146

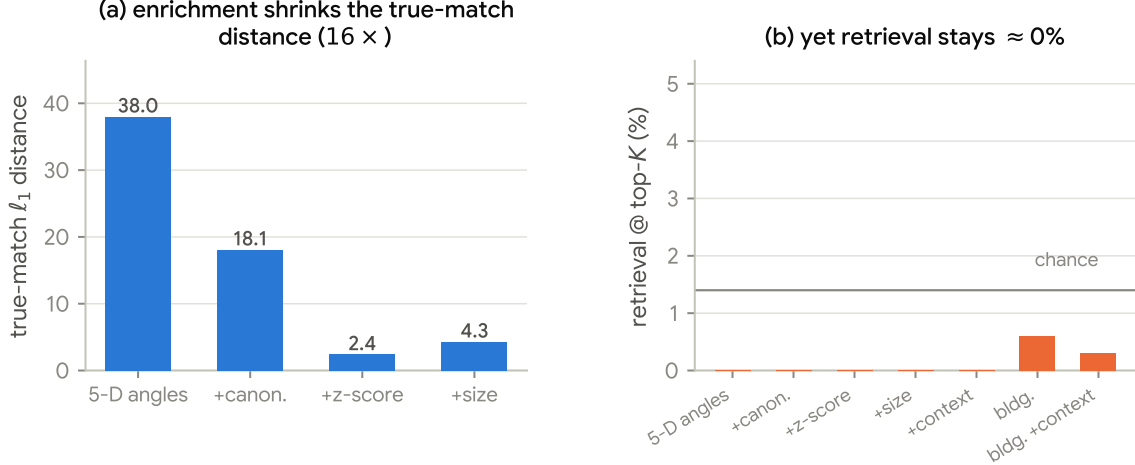


Figure 4: The discriminability ceiling on Changjiang-20 (4.8×10^5 elements, 98 queries). (a) Successive descriptor enrichments shrink the distance between a query triangle and its geometrically correct satellite match by up to $16\times$ (ℓ_1 $38 \rightarrow 2.4$). (b) Yet the fraction of queries for which that correct match is retrieved among the top- K candidates stays at essentially 0% across every variant—below the chance rate expected inside the search window. Shrinking the true-match distance shrinks the impostors’ distances equally; the descriptor is too low-entropy to be unique at scale.

6.6 Impostor density: what actually bounds retrieval

Section 5.4’s basic properties establish that the descriptor is coordinatewise separating (two vectors differ whenever their generating geometry differs) — a fact about vector arithmetic, not about discriminability among real building populations. Here we measure the quantity that does govern discriminability: the density of impostors (other map elements) around the true match’s descriptor value, at the scale of a real city map. For a query triangle and its geometrically correct satellite match at ℓ_1 distance d_{true} , define the impostor count as $N \cdot P(\text{random map element within } d_{\text{true}})$, where N is the map’s total element count; this directly estimates the true match’s expected rank, and hence whether it can appear in a top- K list at all.

Table 6 reports this for all four analysed sites, together with the effective dimensionality of the 5-D descriptor via the PCA participation ratio ($\sum_i \lambda_i^2 / \sum_i \lambda_i$) (a scale-free measure of how many components actually carry variance). Two findings follow. First, the descriptor’s effective dimensionality is 2.0–2.3, confirming quantitatively that most of its variance lives in 2–3 of the 5 nominal dimensions (consistent with e_2, e_3 staying near the constant convex value 180° for the majority of vertices). Second, and decisively, at the true-match distance between 15% and 34% of the *entire* candidate population sits closer than the true match itself (site-dependent: 14.9% on the hilly site to 33.5% on the coastal site), giving an expected true-match rank in the tens to hundreds of thousands—orders of magnitude beyond any practical K . This is the real, measured account of why retrieval collapses to 0% (Table 5): not because the true match is far in descriptor space, but because so many impostors are equally close.

The per-dimension standardization ablation (Table 5) makes this concrete. Standardizing shrinks the true-match ℓ_1 distance by $16\times$ on the urban site ($38 \rightarrow 2.4$); measured independently here across all four sites, standardization shrinks the raw true-match distance by $14\text{--}18\times$ (site-dependent), yet Table 6’s z-scored column shows the expected rank stays essentially *unchanged* on every site (103k vs. 113k on the urban site; comparable on the other three)—because standardization rescales impostor distances by the same factor. Shrinking the true-match distance without shrinking the surrounding impostor density does not improve retrieval; this is the quantitative form of the discriminability ceiling.

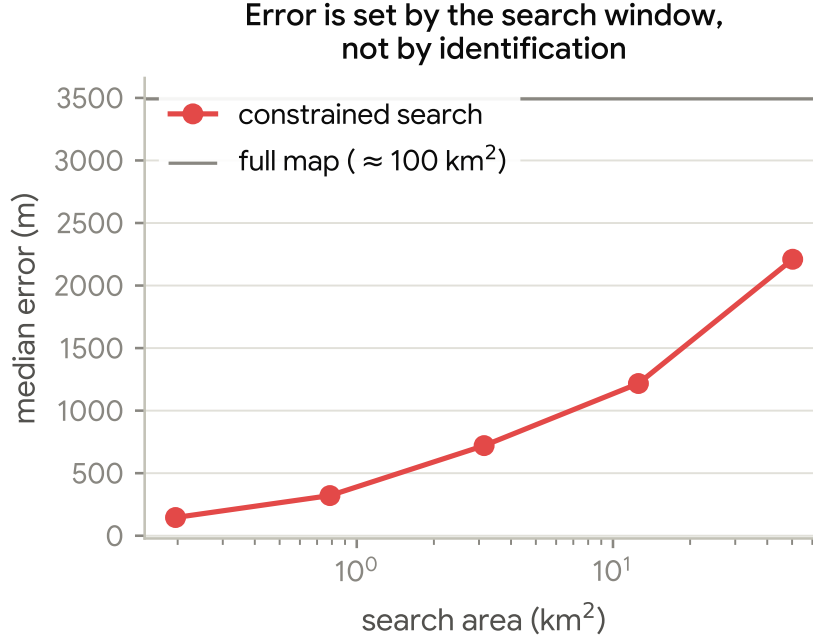


Figure 5: Localization error versus the area of the (coarse-prior) search window. The median error grows monotonically with the search area—from 146 m at 0.2 km² to 2.2 km at 50 km², approaching the full-map error (dashed)—showing that the accuracy in a small window is set by the window bounding the error, not by the descriptor identifying the location.

6.7 Sensitivity to the kernel-expansion depth $D_{\max}$

$D_{\max}$ is fixed to 4 throughout the main results; we ablate it here rather than leave it an untested design choice, sweeping $D_{\max} \in \{0, 1, 2, 4, 8\}$ on both the per-element retrieval diagnostic (sites 01–03) and the area-level recall (all five sites). $D_{\max} = 0$ is the natural control: it disables kernel expansion entirely, so every e_X collapses to the constant convex value 180° and the Ekeland component of the descriptor carries no information beyond a single fixed constant (the two interior-angle dimensions α_1, α_2 are unaffected by $D_{\max}$ and remain informative at every depth).

Table 7 reports both halves. Per-element retrieval stays at 0% from $D_{\max} = 0$ through $D_{\max} = 4$, and is effectively still 0% at $D_{\max} = 8$ (0.03% pooled across the three sites—a single triangle on the coastal site, not a trend). This holds even though the true-match ℓ_1 distance itself *grows* substantially with $D_{\max}$ on the urban site (median 17 → 18 → 22 → 38 → 51 at $D_{\max} = 0, 1, 2, 4, 8$) as more context is folded into each descriptor: a larger, more spread-out true-match distance has no effect on retrieval, reinforcing Section 6.6’s account that distance to the true match is not the binding constraint at any

Table 6: Impostor density at the true-match distance, and PCA effective dimensionality, on the 5-D descriptor (raw and per-dimension standardized). N = candidate map elements; d_{true} = median true-match ℓ_1 distance; $\hat{\rho}$ = participation ratio (effective dim); $E[\text{rank}] = 1 + N \cdot P(\text{impostor within } d_{\text{true}})$.

Site	N	raw		z-scored	
		$\hat{\rho}$	$E[\text{rank}]$	d_{true} (14–18×↓)	$E[\text{rank}]$
01 (urban)	480,569	2.05	113,489	2.21	103,852
02 (coastal)	382,545	2.13	128,228	2.19	103,838
03 (hilly)	1,043,560	2.04	155,655	2.24	166,602
11 (arid)	265,181	2.26	51,662	1.96	54,973

depth—impostor density is. Area-level recall tells a different, and more informative, story: pooled recall rises from 4.0% at $D_{\max} = 0$ (already above the plain baseline’s depth-independent 2.9%, because the interior-angle histogram alone adds signal) through 4.4% at $D_{\max} \in \{1, 2\}$ to a peak of 5.9% at $D_{\max} = 4$, then falls back to 5.1% at $D_{\max} = 8$. $D_{\max} = 4$ is thus not an arbitrary choice: it is close to the empirically-measured optimum, with both less context ($D_{\max} \leq 2$) and more ($D_{\max} = 8$) giving measurably worse pooled recall.

Table 7: Sensitivity to the kernel-expansion depth $D_{\max}$. Per-element retrieval@top-20 (sites 01–03, pooled over $n=6,564$ triangles) stays at 0% throughout, with a negligible 0.03% upturn (one site only) at $D_{\max} = 8$; the area-level pooled recall@top-50 (all five sites, $n=272$ frames) peaks at the paper’s default $D_{\max}=4$. Plain (depth-independent control) is 2.9% throughout.

$D_{\max}$	0	1	2	4	8
Retrieval@top-20 (per-element)	0%	0%	0%	0%	$\approx 0\%$
Area recall@top-50 (pooled, ekeland)	4.0%	4.4%	4.4%	5.9%	5.1%

6.8 Sequential integration does not recover it

Since every accurate map-matching method integrates ambiguous per-frame observations over a trajectory (CFBVM’s Gaussian-mixture matches in a particle filter [29]; SWA-PF’s 4-DoF particle filter [35]), we test whether the same estimator rescues our descriptor. On a 100-frame consecutive flight segment we run a particle filter (2×10^4 particles, random-walk motion of ≈ 266 px/frame matching the measured inter-frame displacement) whose per-frame observation is the Gaussian-mixture of our descriptor’s Hough votes. It does not converge (Table 8, Figure 6): the trajectory-level error stays at kilometre scale (filtered median 5.5 km, *worse* than the single-frame median of 3.2 km, because the filter commits to a spuriously consistent wrong track) and sub-100 m recall is 0%. The diagnostic explains why: the true position receives *any* vote in only 12% of frames (median 0 votes/frame; Figure 6a), and when present its region ranks ≈ 123 rd. A particle filter integrates a signal that *recurs*; ours is absent from 88% of frames, so the filter locks onto a spurious track (Figure 6b). The limiting factor is thus the *observation model*: unlike CFBVM’s area-density descriptor, our per-element geometric descriptor does not provide recurring, spatially consistent support for the true location, which no downstream filter can manufacture. Figure 7 makes this concrete on one frame—the descriptor’s votes scatter across the entire map rather than concentrating at the true position.

Table 8: Two-phase pipeline on a 100-frame Changjiang-20 trajectory. Phase 2 (particle-filter integration of the per-frame Phase 1 matches) does not recover localization, because Phase 1’s observation supports the true position in only 12% of frames.

Estimator	Median error (m)↓
Phase 1 only (single-frame Hough vote)	3214
Phase 1 + Phase 2 (particle filter)	5507
Frames with any support at true position	12%
Sub-100 m recall	0%

Cross-site confirmation (four trajectories, not one). A sequential-integration failure measured on a single 100-frame segment could plausibly be a property of that one trajectory rather than the method. We repeat the identical particle filter on three further 100-frame consecutive segments, one per remaining analysed site (Table 9). The failure mode is consistent across all four: frame-level support for the true position never exceeds 12%, and the filtered trajectory error never drops below kilometre scale. On the coastal site the filter briefly touches sub-100 m error at frame 19 before

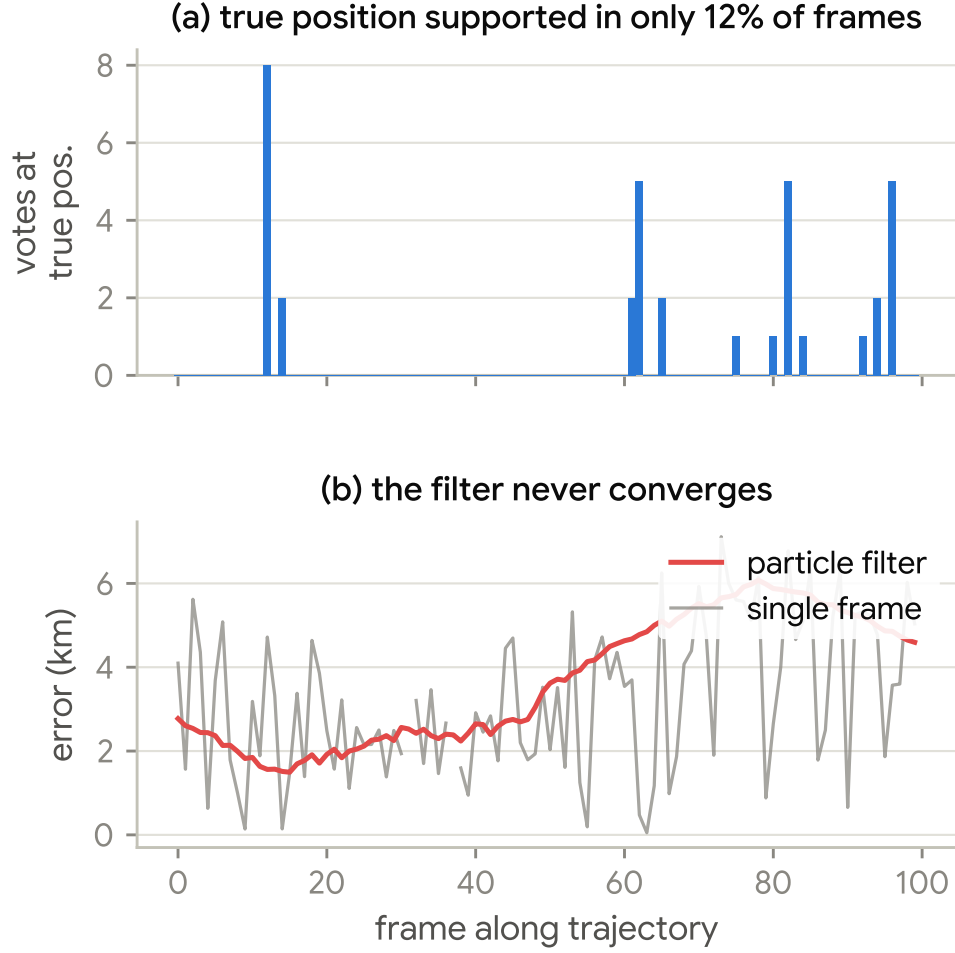


Figure 6: Why sequential integration fails. (a) Votes landing at the true position, per frame along the trajectory: the true location is supported in only 12% of frames (median 0). (b) Position error per frame: the particle filter (red) never converges and drifts onto a wrong track, no better than the raw single-frame estimate (grey). A filter cannot integrate a signal that is absent from 88% of frames.

diverging again—a transient, not a sustained convergence—which we report rather than omit, since it is the kind of near-miss a less careful account would elide.

Figure 8 makes the same point from a different angle, on an independent frame chosen before this figure was designed (the same best-supported frame Figure 7 already uses, so there is no post-hoc selection of a flattering example). For every one of the frame’s 300 query triangles we take its top-100 nearest-descriptor satellite matches and convert each into an implied UAV position, giving a pooled cloud of 30,000 candidate positions. This is a *looser, pooled* statistic than Table 5’s per-triangle retrieval@top- K (which asks whether *each* triangle’s own geometrically correct match is in *its own* top- K): here we only ask how many of all 30,000 candidate votes, pooled across the whole frame, land within a generous 500 px (≈ 150 m) of the true position. Even under this generous, pooled criterion the answer is 170/30,000 (0.6%)—the candidates scatter across the entire map, including geometrically impossible locations such as open water, rather than concentrating near the true position.

Runtime and scale. The pipeline is nonetheless efficient and compact, as intended: descriptor extraction and a single K -NN query take ≈ 0.8 s per image on one CPU core, and the site-01 index

Table 9: Particle-filter integration on four independent 100-frame trajectories (one per site). Frame support and filtered error are consistent across terrains—the sequential-integration failure is a property of the observation model, not of one trajectory.

Site	Frame support	Filtered median err (m)	Sustained < 100 m
01 (urban)	12%	5530	No
02 (coastal)	6%	4901	No (transient touch, frame 19)
03 (hilly)	1%	4640	No
11 (arid)	2%	3758	No



Figure 7: Descriptor votes (orange) for a single, best-supported query frame, overlaid on the Changjiang-20 map; the true UAV position is the green star. The votes scatter across the entire $\approx 100 \text{ km}^2$ map instead of concentrating at the truth—a direct visualization of the discriminability ceiling.

of 4.8×10^5 elements occupies $\approx 12 \text{ MB}$ ($5 \times \text{float32}$ per element). These costs scale linearly in the number of indexed elements; the barrier to city-scale use is accuracy, as analyzed above, not compute or memory.

6.9 Qualitative Visualization

Figure 9 shows Phase 1’s front end on a representative building-dense UAV frame: Mask R-CNN building segmentation (orange outlines) and the interior-filtered CDT mesh (blue) from which the per-triangle descriptors are computed. Segmentation and triangulation are visibly reliable; the failure analyzed in Section 6.5 arises downstream in matching, not here. The matching-stage counterpart is Figure 7 (Section 6.8): the descriptor votes for a query frame scatter across the entire map rather than concentrating at the true position.

Frame 01_0013.JPG: the 100-nearest-descriptor candidates from every query triangle scatter across the map, not toward the true position

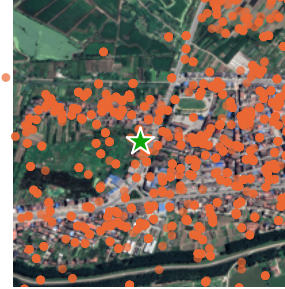
(a) all candidate implied positions



(b) UAV query image



(c) zoom at GT: 170/30000 pooled candidates within 500 px (0.6%)



● top-100 candidates/triangle (30000 total)
★ ground truth

Figure 8: Top-100-per-triangle candidate positions (orange, 30,000 total) vs. ground truth (green star) for one query frame. (a) Candidates cover the whole map, including physically impossible locations (open water). (b) The UAV query image. (c) Zoomed at the true position: only 170/30,000 (0.6%) pooled candidates fall within 500 px—a looser, pooled statistic than the strict per-triangle retrieval@top- K of Table 5, included here for visual intuition, not as a replacement metric.

6.10 Discussion of the failure

The analysis above identifies a single dominant failure mode and rules out several plausible alternatives. It is *not* primarily a coverage problem (the true location is mapped in 93% of cases), nor a segmentation-density problem specific to sparse terrain (the ceiling appears on dense-urban Changjiang), nor a voting or refinement problem (the correct match is absent from the top-20 before any voting), nor an estimator problem (a particle filter cannot converge on it), nor a retrieval-scheme problem (constellation hashing with geometric consensus verification, Section 6.11, fails for the same underlying reason). It is a *discriminability* problem intrinsic to per-element geometric matching: individual building triangles—even enriched with size and neighbourhood context—are not unique among 10^5 – 10^6 candidates, and the discriminative information that identifies a location lives in the joint spatial configuration of many buildings, which per-element retrieval discards. The one regime in which the descriptor localizes is when an external coarse prior shrinks the candidate set (Section 6.5), which is precisely how successful vector-map methods such as CFBVM [29] deploy their matching stage.

6.11 A third line of evidence: constellation hashing also fails, and why

Before pivoting to area-level aggregation, we tested whether a stronger, training-free per-element matcher could recover discriminability directly: astrometry.net-style geometric hashing [21] over building centroids. For every building we form a 4-point “quad” with its three nearest neighbours, compute a similarity-invariant 4-D hash (the two most separated points fixed to $(0, 0)$ – $(1, 1)$; we verified the hash’s invariance and the closed-form similarity-transform recovery on synthetic data, 200/200 trials each), and index all quads in a k -d tree. At query time, each UAV frame’s quads are looked up against this index; each candidate match yields a full similarity transform (translation, rotation, scale) via closed-form least squares, verified by how many of the frame’s *other* buildings project onto a nearby satellite building under that transform.

Segmentation + CDT mesh (01_0001.JPG)



Figure 9: Phase-1 front end on a Changjiang-20 query frame: Mask R-CNN building footprints (orange) and the interior-filtered CDT mesh (blue) that yields the 5-D per-triangle descriptors. Segmentation and triangulation are reliable; matching is the bottleneck (Section 6.5).

This fails too, and the failure is informative. We tested three verification-gate configurations across two sites of contrasting density (dense urban, sparse coastal): unconstrained scale and scale gated near the calibrated value 0.96–1.0 (both on the urban site), and scale and rotation jointly gated near zero, consistent with yaw-aligned frames (on the coastal site, as a cross-site check). Median error stays at 1.9–3.9 km across all three—no better than single-element matching. A direct diagnostic isolates why: even applying the *known, correct* transform (ground-truth position, calibrated scale, zero rotation) and counting how many satellite buildings fall near each projected UAV building, the count is barely distinguishable from applying a *random, wrong* translation at the same scale (mean 4.5 vs. 3.3 buildings within 25 px on the dense site; mean 3.2 vs. 0.6 on the sparse site, where a coincidentally dense reference cluster inflates the null). Two effects compound: cross-view segmentation disagreement means only a minority of buildings visible in a UAV frame have a corresponding detection in the satellite map even at the true location, and building density is high enough everywhere that a wrong transform still lands near *some* buildings by chance. With hundreds of candidate quads verified per frame, the best *wrong* candidate’s building-count regularly matches or exceeds the true candidate’s—geometric verification cannot discriminate a correct correspondence from a lucky coincidence at this density, independent of how the candidates were retrieved. This is a third, independent measurement of the same discriminability ceiling: per-element retrieval, hard theoretical bounds on impostor density (Section 6.6), and geometric consensus verification all hit the same wall from different directions.

6.12 Area-level aggregation recovers discriminability

The diagnosis above is also constructive: if the discriminative signal lives in the joint configuration of many buildings rather than in any single element, then the descriptor should be aggregated over an *area* rather than matched per element. We test this directly while keeping the Ekeland angle central. We tile each map into ≈ 200 m cells and describe every cell by a translation-invariant descriptor: building density, layout moments (centroid spread, mean nearest-neighbour spacing), a footprint-size histogram, and—the shape signal a density field lacks—*histograms of the per-building Ekeland free-cone angles*. Each UAV frame is matched, as a single cell, against the satellite cells; we report coarse-localization recall (fraction of frames whose true cell is in the top-50 of $\sim 10^3$ cells).

To ground this in a genuine head-to-head rather than our own reconstruction, we compare four cell descriptors under *identical* segmentation, cells, and protocol: a *plain* descriptor (density + layout + size, the information a generic area-density field carries); our own re-implementation of **BRM** [4] (a sub-grid building-area coverage ratio); our own re-implementation of **CFBVM**’s [29] concentric-ring building-area profile; and *plain augmented with Ekeland-angle histograms* (ours). BRM and CFBVM here are genuine re-implementations evaluated on our data, not the authors’ own reported numbers (which appear only as landscape context in Table 3).

Statistical treatment. With 70–90 scored frames per site, a single-site recall delta of a few percentage points is not distinguishable from noise, and we do not present it as such. For every comparison we report a bootstrap 95% confidence interval (frame resampling, 10^4 replicates) and a paired permutation test, both per site and pooled across all five sites ($n = 272$ scored frames); an effect is called significant only when the pooled CI excludes zero.

Table 10 and Figure 10 give the full picture, which is genuinely mixed and we report it as such. Area-level matching *escapes the per-element ceiling* on every descriptor: recall rises from the flat 0% of single-element retrieval to several times chance ($\approx 3.7\%$ at top-50) in most cases. Ekeland is the best performer on three of five sites (urban, coastal, hilly); on the built-up site our own *plain* descriptor is best, and on the arid site *both* real baselines outperform Ekeland. Pooled across all five sites, Ekeland significantly outperforms plain (+2.9 pp, 95% CI [0.7, 5.1], $p=0.02$) and significantly outperforms the re-implemented BRM baseline (+4.4 pp, 95% CI [1.5, 7.7], $p\approx 0.012$ – 0.014 by permutation test and McNemar’s test respectively); it is directionally, but not significantly, ahead of the re-implemented CFBVM-rings baseline (+2.6 pp, 95% CI [−1.1, 6.2], $p=0.23$). The pooled effect is significant at top-50 but *not* at top-10 (+0.7 pp, CI [−0.7, 2.2], $p=0.63$): the claim we make is specifically that Ekeland-angle histograms add coarse-localization signal at top-50, not a blanket improvement at every operating

point. The free-cone angle supplies per-building shape information that arrangement- or density-only descriptors (BRM, CFBVM’s regime) do not carry by construction—but the effect is modest, not universal, and a real re-implemented CFBVM baseline is a genuinely close competitor rather than one we clearly surpass.

We are deliberately precise about scope. These are *coarse*-localization gains: absolute recall remains single-digit-to-low-tens of percent, and the top-1 cell is still usually wrong, so the method does not yet deliver precise metric localization. The contribution is a measured, statistically qualified demonstration that (i) the area is the right matching unit for this problem and (ii) the Ekeland free-cone angle is a discriminative, complementary shape cue at that level on most, not all, terrains. Closing the remaining gap to precise localization plausibly requires a learned area embedding or a clean vector map (as used by shape-and-relationship methods on ‘.shp’ e-maps [32]); both are natural extensions of the Ekeland area descriptor introduced here.

Table 10: Area-level matching (≈ 200 m cells, $\sim 10^3$ candidates, 100 frames/site): coarse recall @top-50, four descriptors under identical protocol. BRM and CFBVM-rings are our own re-implementations of the cited methods, evaluated head-to-head—not the original papers’ numbers. Bold = best per site. Bottom rows: pooled ($n=272$) delta vs. Ekeland, 95% bootstrap CI, permutation-test p .

Site	Terrain	Plain	BRM	CFBVM-r.	Ekeland
01	urban	5.5%	0.0%	3.6%	9.1%
02	coastal	2.0%	2.0%	6.1%	10.2%
03	hilly	3.2%	1.6%	0.0%	6.4%
08	built-up	4.3%	0.0%	2.2%	2.2%
11	arid	0.0%	3.4%	5.1%	1.7%
<i>Pooled vs. Ekeland ($n=272$)</i>		+2.9pp	+4.4pp	+2.6pp	—
<i>95% CI</i>		[0.7,5.1]	[1.5,7.7]	[−1.1,6.2]	—
<i>p (permutation)</i>		0.02	0.012	0.23	—
<i>Significant?</i>		Yes	Yes	No	—

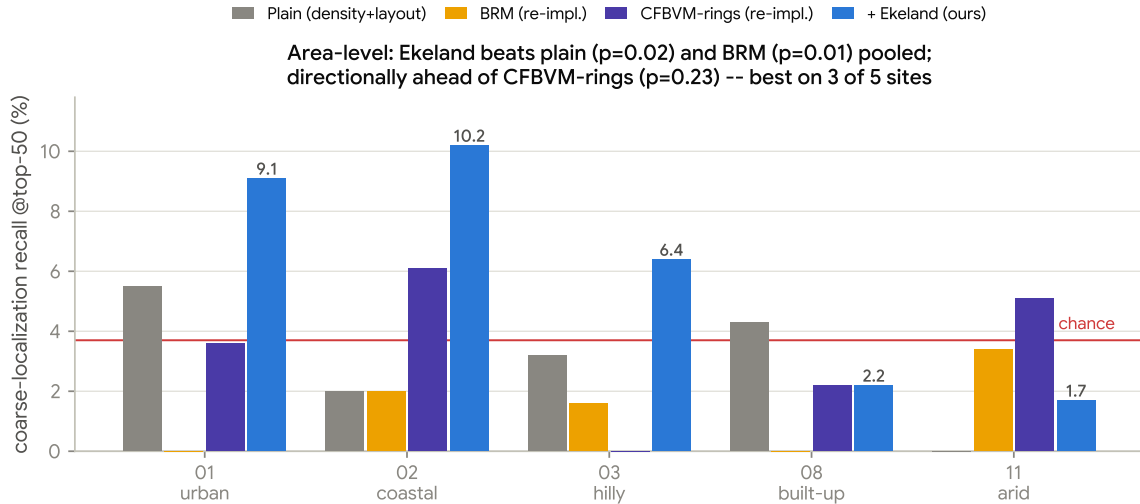


Figure 10: Area-level coarse-localization recall (@top-50) per site, all four descriptors under identical protocol: plain density+layout (grey), the two re-implemented real baselines BRM and CFBVM-rings (yellow, violet), and plain augmented with Ekeland free-cone-angle histograms (ours, blue); horizontal line = chance. Ekeland is the best performer on 3 of 5 sites, not universally—Table 10 reports the pooled bootstrap CIs and significance tests underlying this figure.

7 Conclusion

We have presented a scene-agnostic UAV visual geo-localization framework whose core novelty is a purely geometric 5-D triangle descriptor combining sorted interior angles and edge-anchored Ekeland Free-Cone Angles (MFCA). The key design innovation is the *kernel expansion* P_{sub} : the maximal polygon formed by merging CDT triangles while keeping all three original triangle vertices on the boundary. This stopping criterion is geometrically analogous to Ekeland’s cone-criticality condition, and the resulting Ekeland angle is computed exactly in $O(1)$ per vertex via an interior-angle closed form, replacing the earlier 1° -sweep approximation. The descriptor is invariant to texture, colour, resolution, and season, and requires no scene-specific training. The design of the MFCA component is geometrically inspired by Ekeland’s variational principle [12] (Remark 5.2): each element of the principle (complete metric space, bounded function, cone-touching condition, critical point value) has a natural geometric counterpart in our construction—an analogy that motivates the descriptor’s design and yields a closed-form computation with no sup-not-attained edge case, rather than a literal instantiation of EVP’s analytic machinery. An offline KD-tree index reduces online matching to $O(M \log N)$; plurality voting over satellite patches converts the matching output to a predicted 2-D geographic position.

Our second contribution is an honest, quantitative characterization of this descriptor as a UAV-localization matching stage. On UAV-VisLoc we find that, although building coverage is high (93%), single-frame per-element retrieval never places the correct match in the top-20 and localization error stays at the kilometre scale; that enriching the descriptor with size, a building- constellation shape context, building-level granularity, and dimension normalization shrinks the true-match distance up to $16\times$ without moving retrieval off zero; and that integrating the per-frame matches over a trajectory with a particle filter—the mechanism that makes CFBVM and SWA-PF succeed—also fails to converge, because the true position receives recurring observation support in only 12% of frames. The limiting factor is therefore the *observation model*: purely geometric per-element matching does not, on its own, provide the recurring, spatially consistent support that scale-robust localization requires. This is a useful negative result. It shows that the promising direction is not a richer per-element descriptor but a fundamentally more discriminative *observation*—an area-level or configuration-level descriptor (as in CFBVM’s density rings) or a learned embedding—paired with the coarse-to-fine, sequential estimation that the accurate methods already employ. Our released code and analysis pipeline (`eval.py` and the diagnostic scripts) let future work measure any candidate observation model against the same retrieval, coverage, and sequential-support criteria.

Our third contribution acts on this diagnosis and, crucially, keeps the Ekeland angle central (Section 6.12). Aggregating the descriptor over ≈ 200 m cells—as a histogram of per-building Ekeland free-cone angles alongside building density and layout—escapes the per-element discriminability ceiling: coarse-localization recall rises from a flat 0% to several times chance. Against real re-implemented baselines (BRM, CFBVM’s concentric-ring descriptor) under identical protocol, Ekeland-angle histograms significantly improve pooled coarse recall over plain density/layout (+2.9 pp, $p=0.02$, $n=272$) and over BRM (+4.4 pp, $p\approx 0.012$), and are directionally ahead of CFBVM-rings without reaching significance (+2.6 pp, $p=0.23$). The picture is genuinely mixed per site—Ekeland is best on three of five terrains, plain wins on the built-up site, and both real baselines beat Ekeland on the arid site—which is what makes the pooled, statistically qualified claim credible rather than a cherry-picked win. This isolates the Ekeland free-cone angle as a discriminative shape cue at the area level that arrangement- or density-only descriptors do not by construction carry, on most but not all terrains, and points to the concrete next step: a learned area embedding, or matching against clean vector maps [32], built on the Ekeland area descriptor introduced here.

Limitations and scope. (i) Our multi-site evaluation covers five of the eleven UAV-VisLoc sites, selected for terrain diversity (urban, coastal, hilly, built-up, arid); the discriminability ceiling was consistent across all five, and the area-level Ekeland gain is significant when pooled but is not the best performer on every individual site (Section 6.12); the remaining six sites and the full per-site error distributions are left to future work. (ii) Our area-level result is a coarse-localization study (top- K

cell recall among $\sim 10^3$ cells); we do not yet demonstrate precise metric localization, which we expect requires either a learned area embedding or a clean vector map, and did not implement. (iii) The method assumes approximately nadir views, buildings in the field of view, and heading to $\pm 5^\circ$; open terrain without buildings is out of scope. (iv) SWA-PF [35] is, as of this writing, an arXiv preprint with no peer-reviewed venue found; its numbers are reproduced as reported and should be re-checked against a peer-reviewed version if one appears.

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